

LoRa RFM9x module connected to Arduino Uno

lostresearcher

Feb 2022 post #1

Newbie here. I'm trying to connect the RFM9x LoRa to Arduino Uno to make it work. I followed the everything in the adafruit tutorial and used the code in this link [RFM9X Test | Adafruit RFM69HCW and RFM9X LoRa Packet Radio Breakouts | Adafruit Learning System](#). However I'm getting this error (LoRa radio init failed) . May I humbly ask for any recommendations I could take on this matter? Thanks in advance...

sterretje

Feb 2022 post #2

@lostresearcher , your topic has been moved to a more suitable location on the forum. *Installation and Troubleshooting* is not for problems with (nor for advise on) your project 😊 See [About the Installation & Troubleshooting category](#).

Please post **your** code here (and, just in case, don't forget to use code tags). Please post a wiring diagram of **your** setup; a clear photo in daylight might also be helpful.

lostresearcher

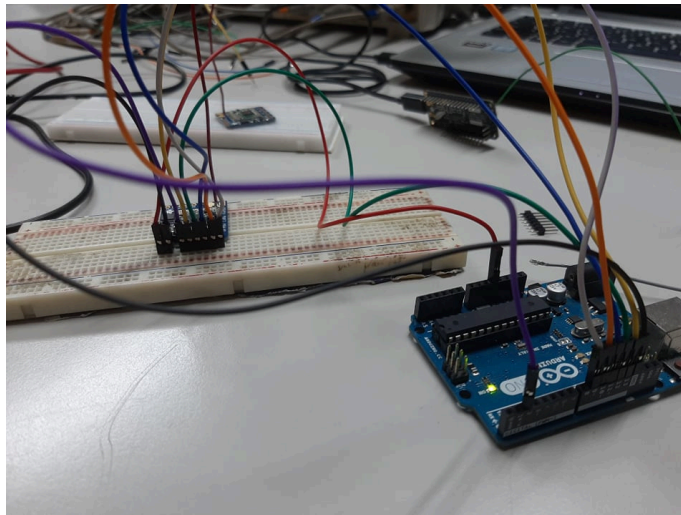
Feb 2022 post #3

Thank you @sterretje 😊 I am not aware of the categories since I am kinda new here.

My wiring goes like this...

Arduino Uno ----- RFM9x Lora Module (434 MHz)

5V ----- Vin
GND ----- GND
D2 ----- GO (interrupt)
D13 ----- SCK
D12 ----- MISO
D11 ----- MOSI
D10 ----- CS
D9 ----- RST



Here's the code I'm trying. I copied them based on adafruit's RFM9x Lora Module tutorial.

Transmitter's Code

```
// LoRa 9x_TX
// -*- mode: C++ -*-
// Example sketch showing how to create a simple messaging client (transmitter)
// with the RH_RF95 class. RH_RF95 class does not provide for addressing or
// reliability, so you should only use RH_RF95 if you do not need the higher
// level messaging abilities.
// It is designed to work with the other example LoRa9x_RX

#include <SPI.h>
```

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```

#include <RH_RF95.h>

#define RFM95_CS 10
#define RFM95_RST 9
#define RFM95_INT 2

// Change to 434.0 or other frequency, must match RX's freq!
#define RF95_FREQ 434.0

// Singleton instance of the radio driver
RH_RF95 rf95(RFM95_CS, RFM95_INT);

void setup()
{
  pinMode(RFM95_RST, OUTPUT);
  digitalWrite(RFM95_RST, HIGH);

  while (!Serial);
  Serial.begin(9600);
  delay(100);

  Serial.println("Arduino LoRa TX Test!");

  // manual reset
  digitalWrite(RFM95_RST, LOW);
  delay(10);
  digitalWrite(RFM95_RST, HIGH);
  delay(10);

  while (!rf95.init()) {
    Serial.println("LoRa radio init failed");
    while (1);
  }
  Serial.println("LoRa radio init OK!");

  // Defaults after init are 434.0MHz, modulation GFSK_Rb250Fd250, +13dBm
  if (!rf95.setFrequency(RF95_FREQ)) {
    Serial.println("setFrequency failed");
    while (1);
  }
  Serial.print("Set Freq to: "); Serial.println(RF95_FREQ);

  // Defaults after init are 434.0MHz, 13dBm, Bw = 125 kHz, Cr = 4/5, Sf = 128kbs
  // The default transmitter power is 13dBm, using PA_BOOST.
  // If you are using RFM95/96/97/98 modules which uses the PA_BOOST transmitter
  // you can set transmitter powers from 5 to 23 dBm:
  rf95.setTxPower(23, false);
}

int16_t packetnum = 0; // packet counter, we increment per xmission

void loop()
{
  Serial.println("Sending to rf95_server");
  // Send a message to rf95_server

  char radiopacket[20] = "Hello World #      ";
  itoa(packetnum++, radiopacket+13, 10);
  Serial.print("Sending "); Serial.println(radiopacket);
  radiopacket[19] = 0;

  Serial.println("Sending..."); delay(10);
  rf95.send((uint8_t *)radiopacket, 20);

  Serial.println("Waiting for packet to complete..."); delay(10);
  rf95.waitPacketSent();
  // Now wait for a reply
  uint8_t buf[RH_RF95_MAX_MESSAGE_LEN];
  uint8_t len = sizeof(buf);

  Serial.println("Waiting for reply..."); delay(10);
  if (rf95.waitAvailableTimeout(1000))
  {
    // Should be a reply message for us now
    if (rf95.recv(buf, &len))
    {
      Serial.print("Got reply: ");
      Serial.println((char*)buf);
      Serial.print("RSSI: ");
      Serial.println(rf95.lastRssi(), DEC);
    }
  }
  else

```

```

    {
        Serial.println("Receive failed");
    }
}
else
{
    Serial.println("No reply, is there a listener around?");
}
delay(1000);
}

```

Receiver's Code

```

// Arduino9x_RX
// -*- mode: C++ -*-
// Example sketch showing how to create a simple messaging client (receiver)
// with the RH_RF95 class. RH_RF95 class does not provide for addressing or
// reliability, so you should only use RH_RF95 if you do not need the higher
// level messaging abilities.
// It is designed to work with the other example Arduino9x_TX

#include <SPI.h>
#include <RH_RF95.h>

#define RFM95_CS 10
#define RFM95_RST 9
#define RFM95_INT 2

// Change to 434.0 or other frequency, must match RX's freq!
#define RF95_FREQ 434.0

// Singleton instance of the radio driver
RH_RF95 rf95(RFM95_CS, RFM95_INT);

// Blinky on receipt
#define LED 13

void setup()
{
    pinMode(LED, OUTPUT);
    pinMode(RFM95_RST, OUTPUT);
    digitalWrite(RFM95_RST, HIGH);

    while (!Serial);
    Serial.begin(9600);
    delay(100);

    Serial.println("Arduino LoRa RX Test!");

    // manual reset
    digitalWrite(RFM95_RST, LOW);
    delay(10);
    digitalWrite(RFM95_RST, HIGH);
    delay(10);

    while (!rf95.init()) {
        Serial.println("LoRa radio init failed");
        while (1);
    }
    Serial.println("LoRa radio init OK!");

    // Defaults after init are 434.0MHz, modulation GFSK_Rb250Fd250, +13dBm
    if (!rf95.setFrequency(RF95_FREQ)) {
        Serial.println("setFrequency failed");
        while (1);
    }
    Serial.print("Set Freq to: "); Serial.println(RF95_FREQ);

    // Defaults after init are 434.0MHz, 13dBm, Bw = 125 kHz, Cr = 4/5, Sf = 128khi
    // The default transmitter power is 13dBm, using PA_BOOST.
    // If you are using RFM95/96/97/98 modules which uses the PA_BOOST transmitter
    // you can set transmitter powers from 5 to 23 dBm:
    rf95.setTxPower(23, false);
}

void loop()
{
    if (rf95.available())
    {

```

```
// Should be a message for us now
uint8_t buf[RH_RF95_MAX_MESSAGE_LEN];
uint8_t len = sizeof(buf);

if (rf95.recv(buf, &len))
{
    digitalWrite(LED, HIGH);
    RH_RF95::printBuffer("Received: ", buf, len);
    Serial.print("Got: ");
    Serial.println((char*)buf);
    Serial.print("RSSI: ");
    Serial.println(rf95.lastRssi(), DEC);

    // Send a reply
    uint8_t data[] = "And hello back to you";
    rf95.send(data, sizeof(data));
    rf95.waitPacketSent();
    Serial.println("Sent a reply");
    digitalWrite(LED, LOW);
}
else
{
    Serial.println("Receive failed");
}
}
```

sterretje**Feb 2022 post #4**

Please edit your post, select all code and apply code tags using the `</>` button; next save your post.



Type here. Use Markdown, BBCode, or HTML to format. Drag or paste images.

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  Adafruit RFM96W LoRa Radio init failed	6	Sep 2020